

Homegrown Restaurant Food Menu Pricing System

Project Step 4 Draft Version
Project Proposal, Outline, ERD,
Normalized Schema, Sample Data,
UI Interface, DML, DDL, PL SQL,
RESET and SELECT

CS 340

Summer 2026

8/6/2026

Live Application Link: <http://flip1.engr.oregonstate.edu:46231/>

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I Feedback by the TAs and Peer Reviewers

1.1 Peer review by Logan Jordan on Step 1 Draft

“Does the overview describe what problem is to be solved by a website with DB back end?”

Yes. The overview explains that Homegrown needs to calculate the cost of each menu item in order to determine the profit margins. It says a DB backend will track ingredient costs and vendor pricing to help solve this.

Does the overview list specific facts?

The only number given is the annual revenue of more than \$1 million. I would suggest to add some other metrics like the number of menu items, ingredients, or vendors to better show the scale of the system.

Are at least four entities described, and does each one represent a single idea to be stored as a list?

Yes. There are MenuItem, Ingredient, Vendor, MenuItemIngredient, and VendorIngredient, each represent a single idea and stored in its own table.

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities?

VendorIngredient is missing a purpose statement. Ingredient in the ERD has a preferredVendorID attribute that isn't mentioned in the outline.

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?

Yes. The 1:M relationships are correct. The M:N relationships MenuItem to Ingredient and Ingredient to Vendor are implemented correctly with intersection tables. The ERD is logical overall.

Is there consistency in a) naming between overview and entity/attributes, b) entities plural, attributes singular c) use of capitalization for naming?

All the entities are plural and attributes are singular, however capitalization is inconsistent. MenuItem attributes are snake_case, like menu_itemID, while Ingredient and Vendor attributes are camelCase such as ingredientID and vendorID. I would suggest to pick one format and use it for all the tables.

This review is my own work and original.”

1.2 Peer review by Yusuf Morsi on Step 1 Draft

“Hello Group 41!

I've just got done going over your project step 1 draft, and I admire your choice of using a real world scenario for your project idea, as our group chose to go in a similar direction by opting to build a utility billing management system for our project. I observed your draft included five quality and clear entities, including MenuItem, Ingredient, MenuItemIngredient, Vendor, and VendorIngredient, with each entity including quality attributes that match what's expected under each of the chosen entities. I also

observed you comprehensively covered all the required relationships with four Many-to-Many, four One-to-Many, and four Many-to-One relationships spread between your five entity's. Your project draft also clearly explains the problem it'll be targeting and hoping to solve, and again, I applaud your group for selecting a real world scenario to build your project on! I looked over your ER diagram, and it seems to match your database outline well, but I believe it could be made a bit clearer. Overall, great work on creating a comprehensive and well made first draft for your project, and I wish your group the best of luck on the rest of this course!"

1.3 Peer review by Gabriel Venegas on Step 1 Draft

"Hey Anton and Borislava, awesome project idea! Tracking restaurant to help family business fix menu prices is super smart and useful. Just a couple of quick tips to make it perfect: try to stick to being consistent w/ camelCase instead of mixing it with snake_case. Also you can probably delete that dashed line connecting Vendors and Ingredients directly on your diagram, since your 'VendorIngredients' intersection table already handles that connection. Can i suggest you switch 'vendorContact' from an "int" to a "VARCHAR". Which will keep MySQL from throwing errors on long phone numbers later. Overall great job altogether"

1.4 Feedback from grader Saranya Sounder Rajan

"Explicitly use the entity names. On your ERD, I am unsure as to why there's an extra 1:M relationship between vendors and ingredients."

1.5 Feedback from grader Saranya Sounder Rajan on Step 2 Draft

"Dear team,

Overall, great work so far. A couple of quick notes:

Schema: Please revisit the schema module and try redoing your schema using phpMyAdmin.

Ingredients table: Does your Ingredients table need this many sample rows in your data dump? Consider trimming it down for clarity."

1.6 Peer review by George Cross on Step 2 Draft

"Hello Group 41,

I have reviewed your database project overview and database definition. For clarity and brevity, I have pasted my responses to each of the Project Step 2 review questions below:

- **Q: Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?**
 - A: Yes, the Schema diagram follows the database outline, but it does not use standard crow's-foot notation for entity relationships. I suggest reviewing the requirements for Project Step 2 and verifying that you have met the requirements for the Schema diagram stated.
- **Q: Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**
 - A: Yes, the naming of the entities and relationships are consistent across the outline, ER diagram, and Schema diagram. However, the Sample Data has an error in the naming of the "Vendor Ingredients" table.

- **Q: Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?**
 - A: No, the Schema is not easy to read or clear as the format does not follow crow-foot notation and the arrows don't clearly illustrate the relationships. As an example, "Ingredients" points directly to "Vendors" without going through any intersection table.
 - **Q: Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?**
 - A: Yes, intersection tables are properly formed, with the exceptions noted in the previous response above.
 - **Q: Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?**
 - A: I don't think there is a normalization/dependency issue with your data, but if you want to have more than one vendor listed for an available ingredient more efficiently (3NF), then you might consider adding the determination of whether the Vendor is preferred within the intersection table "VendorIngredients". This way, if a Vendor changes their sale price then the vendor and intersection table can update automatically for the lowest price and then send the preferred vendorID automatically to the ingredients table. Otherwise, ingredients has to look up unit price in the intersection table to determine the preferred vendor each time. I believe this makes tracking preferred vendors easier/more efficient as the data for preference is already stored in the intersection table. I am fairly inexperienced with database formats so I could be overcomplicating this, but thought it was worth mentioning for consideration.
 - **Q: Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)**
 - A: While the SQL file looks syntactically correct, your group has added excessive sample data rows (>10 entries) which makes verifying the accuracy and integrity of the data more difficult for reviewers (30 ingredients in your sample data table, when the limit stated in the assignment page is 10 max per table). I suggest only adding the requested amount of data so that it is easier for a reviewer to validate data integrity, and this will also more accurately meet the assignment rubric requirements.
 - **Q: In the SQL, are the data types appropriate considering the description of the attribute in the database outline?**
 - A: Yes, all datatypes are appropriate for the attributes.
 - **Q: In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?**
 - A: Primary Keys and foreign keys are correctly defined, and Cascade operations are declared. Nice job!
 - **Q: In the SQL, are relationship tables present when compared to the ERD/Schema?**
 - A: Yes, all relationship tables are present, with Foreign Keys relationships defined.
 - **Q: In the SQL, is all example data shown in the PDF INSERTED?**
 - A: Yes, however, I strongly suggest limiting your sample data to be within the parameters given in the assignment. Right now, your sample data contains 30 ingredients and makes data integrity validation excessively cumbersome.
 - **Q: Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?**
 - A: Yes, the SQL file is well structured and commented sufficiently.
- Statement of Originality:** All responses to the CS340 Project Step 2 review questions stated above are entirely my own. The question prompts were taken from Oregon State

University's CS340 (CS_340_400_U2026) Project Step 2 Review assignment page on Canvas.

Work Cited: Oregon State University. (n.d.). CS 340 - *Project Step 2 Review* [Canvas assignment]. Canvas. Retrieved July 27, 2026, from https://canvas.oregonstate.edu/courses/2051721/assignments/10565920?module_item_id=26923346

Best!

-George Cross”

1.7 Peer review by Caleb Barackman on Step 2 Draft

“Not 100%. It is mostly correct, but the data types do not all match with the outline. For example, ingredientName in the outline is VARCHAR(155) but in the schema it is VARCHAR(255). unitType in the outline is VARCHAR(4) but in the schema it is VARCHAR(55).

- All naming is matched between outline, ER, and schema. Capitalization is consistent and text format is the same.
- The schema is easy to read. Could be slightly improved by making the line from VendorIngredients to Ingredients a single 90 degree angle.
- Intersection tables are properly formed, both with a PK and two FKs.
- Sample data looks good, with nothing that would go against 1NF, 2NF, 3NF.
- SQL file is syntactically correct.
- Data types match the schema diagram exactly. Because the schema and outline are not consistent with data types, it does not match the outline.
- Primary and Foreign keys are present and done correctly to match the schema. Cascade operations are correct.
- Relationship tables are present like in the schema.
- All example data is inserted correctly in the SQL.
- The SQL is well structured and easy to read. Comments are present describing the tables creation as well as reasons for the cascade options chosen.
- All review work is my own with no AI used.”

1.8 Peer review by Nathan Anderson on Step 2 Draft

“Hello Group 41,

This is a very creative idea, and a great use of the class project!

- Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

No the outline uses specifics like varchar(155) and DECIMAL(4,2) while the schema itself uses varchar(255) and DECIMAL(10,2). This may just be intentional to make the schema simpler, but worth noting in case. DDL matches schema.

- Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

Naming matches except in the ddl menuItemId doesn't match since the rest use ID. Simple fix. Entities plural are correct and match, naming is consistent.

- Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes the schema is easy to read and understand. Relationships are easy to read and understand, no lines are crossing.

- Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes both are formed properly and facilitate the correct M:N relationship.

- Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

There is no transitive dependencies or partialized dependencies issues.

- Is the SQL file syntactically correct? This can be easily verified by using PhpMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Upon running this on my own database, I had no issues. Everything worked properly so this should be correct.

- In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

Yes all data types are appropriate and match what the database is trying to accomplish.

- In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

All the primary and foreign keys are correctly defined between the SQL and Schema. Appropriate CASCADE operations are present.

- In the SQL, are relationship tables present when compared to the ERD/Schema?

Yes, the relationships are all mapped correctly compared to the schema.

- In the SQL, is all example data shown in the PDF INSERTED?

Yes all the example data used in the SQL is shown in the pdf correctly.

- Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

Yes the SQL is well commented and structured.

This work is all my own."

1.9 Peer review by Ahmed Abdulmohsen A Almazroa on Step 2 Draft

"Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

Yes, just a few comments

- vendorContact: Database Outline correctly shows VARCHAR(13). But the Normalized Schema still shows vendorContact: VARCHAR(255)

- unitType length: In Outline VARCHAR(4), while in schema VARCHAR(55)

- Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

Yes, there is there consistency in naming, entities plural, attributes singular and use of capitalization for naming?

- Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes, it is clear and easy to read

- Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes, intersection tables properly formed

- Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

No partial or transitive dependencies issues

- Is the SQL file syntactically correct? This can be easily verified by using PhpMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

I run the code using PhpMyAdmin and had no issues.

- In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

Data types are mostly appropriate and match the outline, but vendorContact VARCHAR(255) contradicts the outline's stated VARCHAR(13) as aforementioned.

- In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

Yes, PK and FK definitions are all correctly defined against the schema, and CASCADE choices are appropriate and match the relationships described in the ERD and outline.

- In the SQL are relationship tables present when compared to the ERD/Schema?

Yes, both relationship tables required by the ERD and Schema are present in the SQL,

- In the SQL, is all example data shown in the PDF INSERTED?

Yes, all example data is inserted

- Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

Yes, it is well organized structured

- All work is my own, I used AI for language clarification"

1.10 Peer review by Alec Moschetti on Step 3 Draft

"Hi Anton and Bobbi,

You've got SELECT views for all the entities and M:M junctions in your schema and all of your ADD, UPDATE, and DELETE forms are on the same view pages. I will note that some of the form fields for your UPDATES have some mismatches between your DML.sql queries and what form fields exist on the page. For example, the manage menu items page has no way of updating the menu item cost through an input but in your DML.sql there is clearly an update to the cost SQL statement. The same mismatch exists for the ingredients update inputs as well. A slight discrepancy which is not a big deal for this step of the project, but just thought I would point that out. Otherwise yes all your views have the proper UI for the SELECT, INSERT, UPDATE, and DELETE for M:M and individual entities from the schema.

Your column names in your tables in all of your views have nicely formatted and logical names which is great. One thing to note is that you have a style discrepancy across the pages. The home, menu items, ingredients pages are all styled one way (or possibly not styled?) and the vendors and the two M:M junction pages have a different set of styles applied. The name of the navigation items even change as

well between those two sets of views which is a little jarring from a UI perspective. Not ultimately a big deal for this step but something that I could mention anyhow.

Overall nice job getting your site live on the school server and having all the requirements met.

My review was entirely my own thoughts and work, no outside sources were used."

1.11 Peer review by Kyle Sydness on Step 3 Draft

"Hi Anton and Bobbi, the content on your UI is all excellent, so I don't really have anything suggestions to offer in that regard. My only suggestions are cosmetic.

All tables from the schema are represented through a select query in the UI.

There is an insert/add option available for all tables in the schema.

There are delete options for all tables in the UI. There are two different delete options on the pages - one where there is a button in the row and one where there is a form below requiring the row ID. I'd suggest deciding which you'd like to stick with (I think either would be good to stick with).

For the M:M relationships, all the rows are set to delete their rows in the intersection tables without interfering with the other table.

There are options to update entities for all of the tables in the UI.

Is there at least one UPDATE form in the UI to modify an M:M relationship? In other words, does the UPDATE allow the user to select a different foreign key value to update the intersection table with?

For the Intersection tables, the update option correctly shows options based on the FKs for the M:M relationships.

If you have the chance, I'd suggest updating the style and color. Right now it's just simple black and white.

There's also a discrepancy in style for some of the pages, which I've already mentioned, so I'd suggest codifying the different pages to keep consistency.

All work is my own"

1.12 Peer review by Arjun Rahul Bhawe on Step 3 Draft

"Does the UI utilize a SELECT for every table in the schema? Based on the DML.sql, yes there's a dedicated SELECT for each of the 5 tables (MenuItems, Ingredients, Vendors, MenuItemIngredients, VendorIngredients), and the two intersection-table SELECTs correctly JOIN out to show names instead of raw IDs. Good use of AS aliases too (preferredVendor, etc.), which answers your own "other suggestions" prompt before I could even ask it. One thing I can't confirm: the draft PDF doesn't include any UI screenshots, so I'm going off the DML/DDDL alone. I wasn't able to reach the live link (classwork.engr.oregonstate.edu:46231) to see the pages rendered. I'd strongly suggest adding screenshots of each page in the next draft so reviewers/graders don't have to be on the OSU VPN to verify the UI actually matches these queries.

Does the UI implement an INSERT for at least one table? Yes, every table has an INSERT statement with clearly named @xInput placeholders, including the two intersection tables.

Does the UI have at least one DELETE for any entity? Yes, all 5 tables have DELETE statements.

Does the UI have at least one DELETE that removes rows from a M:M relationship (without deleting the parent entities)? Yes, and this is done well, MenuItemIngredients and VendorIngredients both use ON DELETE CASCADE on their FKs, so deleting a MenuItem or Ingredient correctly cleans up the intersection rows without touching the other parent table (e.g., deleting an Ingredient removes its MenuItemIngredients/VendorIngredients rows but doesn't delete any MenuItems or Vendors). This is a correct implementation of the requirement.

Is there at least one UPDATE form for a single entity? Yes, all 5 tables support UPDATE.

Is there at least one UPDATE that modifies an M:M relationship (changes a FK)? Yes, both MenuItemIngredients and VendorIngredients UPDATE statements let you reassign the FK values from a dropdown (menuItemID_from_dropdown_Input, ingredientID_from_dropdown_Input, etc.), not just edit the non-key attribute.

Other suggestions:

Biggest thing: add UI screenshots to the draft PDF. The write-up in section III says a full Node/Express/Handlebars front end exists for all 5 tables, and the DML strongly backs that up, but a reviewer/grader shouldn't have to trust the description or be on VPN to confirm it.

menuItemTotalCost DECIMAL(4,2) caps out at 99.99 fine for a single item, but if this ever needs to hold a combo/catering total it'll overflow. Not a blocker, just worth a second look.

Consider a CHECK (ingredientAmount > 0) / CHECK (unitCost > 0) on the intersection tables, similar to the unitType CHECK you already added keeps bad data out at the DB level rather than relying on the frontend form.

The preferredVendorID ON DELETE SET NULL is a nice touch and matches the outline's stated behavior.

All work is my own."

1.12 Peer review by Alexandra Meyers on Step 3 Draft

"Does the UI utilize a SELECT for every table in the schema?

Each table is displayed on each page.

Does the UI implement an INSERT form for at least one table in the schema?

There is at least one insert form.

Does the UI have at least one DELETE for any one entity?

There is at least one delete.

Does the UI have at least one DELETE that will remove things from a M:M relationship?

Yes.

Is there at least one UPDATE form in the UI for any one entity?

Yes

Is there at least one UPDATE form in the UI to modify an M:M relationship?

Yes

Do you have any other suggestions for the team to help with their HTML UI? For example, using AS aliases to replace obscure column names such as fname with First Name.

I would suggest having a drop down list for your menu items page for deleting and updating your menu. Right now, you have the end user writing in the Menu item ID, but if you end up with more than three items, it will be hard to keep track of all the menuItemIDs. Other than that, all of your pages are clean and easy to read! Great job.

This work is all my own.”

1.13 Feedback from grader Saranya Sounder Rajan on Step 2 Draft

“Dear team,

Overall, great work so far. A couple of quick notes:

Schema: Please revisit the schema module and try redoing your schema using phpMyAdmin.

Ingredients table: Does your Ingredients table need this many sample rows in your data dump? Consider trimming it down for clarity.”

II Actions Based on the Feedback

- Added metrics for number of menu items, cost of goods sold (COGS) recommended percentage and an example of the difference in price for ingredients between vendors.
- Included a purpose statement for the VendorIngredients entity.
- Changed the datatype for the attribute vendorContact to VARCHAR(13), so it can handle various phone number inputs.
- Changed menu_itemID, menu_itemName, and menu_itemTotalCost to camelCase to keep consistency.
- Edited ERD to remove direct relationship line between Vendors and Ingredients that represented the M:N “vendors sell ingredients” relationship since VendorIngredients intersection table already models that connection
- Edited ERD to make it clearer/ cleaner so that it’s easier to interpret
- Added preferredVendorID as a proper FK to match with the ERD
- Clarified the 1:M relationship between Vendors and Ingredients through the preferredVendorId foreign key.

Actions Based on Feedback for Draft 2:

- Recreated Schema Diagram so it uses standard crow’s-foot notation and so that it clearly shows every M:N relationship going through the intersection tables. Also

the only direct line between ingredients and Vendors in the 1:M preferred VendorID foreign key relationship which is now labeled

- Shortened the Ingredients sample data from 30 to 10 rows (MenuItemIngredients and VendorIngredients were also shortened to only reference the remaining ingredients)
- Updated DDL and Schema so that every data type matches database outline
- Renamed columns so that all IDs consistently end in ID (menuItemID, menuItemIngredientID, vendorIngredientID)

Actions Based on Feedback for Draft 3:

- Edited pages to keep consistency across all the different pages

III Upgrades to the Draft Version

- Added web application front end ([Node.js/Express](#) application using Handlebars templating engine. Has UI page for each of the five tables: MenuItems, Ingredientes, Vendors, MenuItemIngredients, VendorIngredients).
- Added PL.SQL with demonstration CUD operation
- SELECT implemented on all five entity pages
- Expanded sample data

IV Overview

Homegrown is a small, family-owned restaurant in Florence, OR, with annual revenue from food sales exceeding \$1 million. One of the largest expenses in a restaurant is the cost of goods sold (COGS). An industry standard for sustainable operations is for COGS to not exceed 30-35% of the menu price for a particular item. The Homegrown menu includes approximately 40 food items, of which 10-12 are best sellers. When pricing these core items, a seemingly small difference in ingredient price can largely affect the overall average COGS percentage. Similarly, an ingredient which is used in multiple menu items has a higher influence on pricing. For example, a 30 lb case of French fries may be \$49 at US Foods and \$41 at Sysco. While the difference per serving is only \$0.13, it adds up to several hundred dollars per month.

A website with a database backend would facilitate the accurate calculation of the cost of each menu item. This information can then be used to set the menu prices at the desired profit margin for each item.

V Database Outline

- **MenuItems:** records the details of each item on the food menu.
 - menuItemID: int(10), auto_increment, unique, not NULL, PK
 - menuItemName: VARCHAR (155), not NULL
 - menuItemTotalCost: decimal(4,2), not NULL
 - relationship:
 - M:N with the entity Ingredients - multiple menu items can be made of multiple ingredients.
 - 1:M with the entity MenuItemIngredients, which serves as an intersection table.

- **Ingredients:** records the details of all food ingredients.
 - ingredientID: int(10), auto_increment, unique, not NULL, PK
 - ingredientName: VARCHAR(155), not NULL
 - unitType: VARCHAR(4), not NULL (constraint: only accepts “oz” or “each” as the two possible measurements)
 - preferredVendorID: int(10), FK
 - Relationship:
 - M:N with the entity MenuItems - multiple ingredients can be in multiple menu items.
 - 1:N with the entity MenuItemIngredients, which serves as an intersection table.
 - M:N with the entity Vendors - multiple ingredients can be sold by multiple vendors.
 - 1:N with the entity VendorIngredients, which serves as an intersection table.
 - M:1 with the entity Vendors through the preferredVendorId foreign key, which identifies a preferred vendor for each ingredient based on things such as cost, quality and availability (each ingredient can have at most 1 preferred vendor at a time. A vendor can be preferred for many ingredients.)

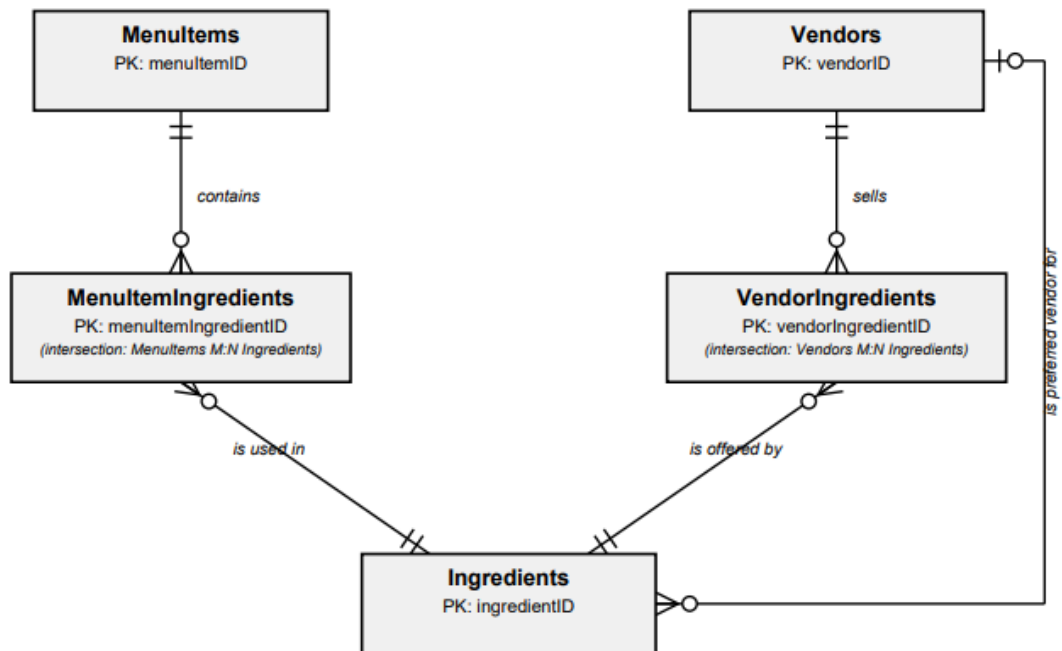
- **MenuItemIngredients:** records which ingredients are used to prepare a menu item, and the amount of each used.
 - menuItemIngredientID: int(10), auto_increment, unique, not NULL, PK
 - menuItemID: FK
 - ingredientID: FK
 - ingredientAmount: decimal(4,2), not NULL
 - relationship:

- M:1 with the entity MenuItem (many ingredients go into a single menu item)
 - N:1 with the entity Ingredients (one menu item can have multiple ingredients)

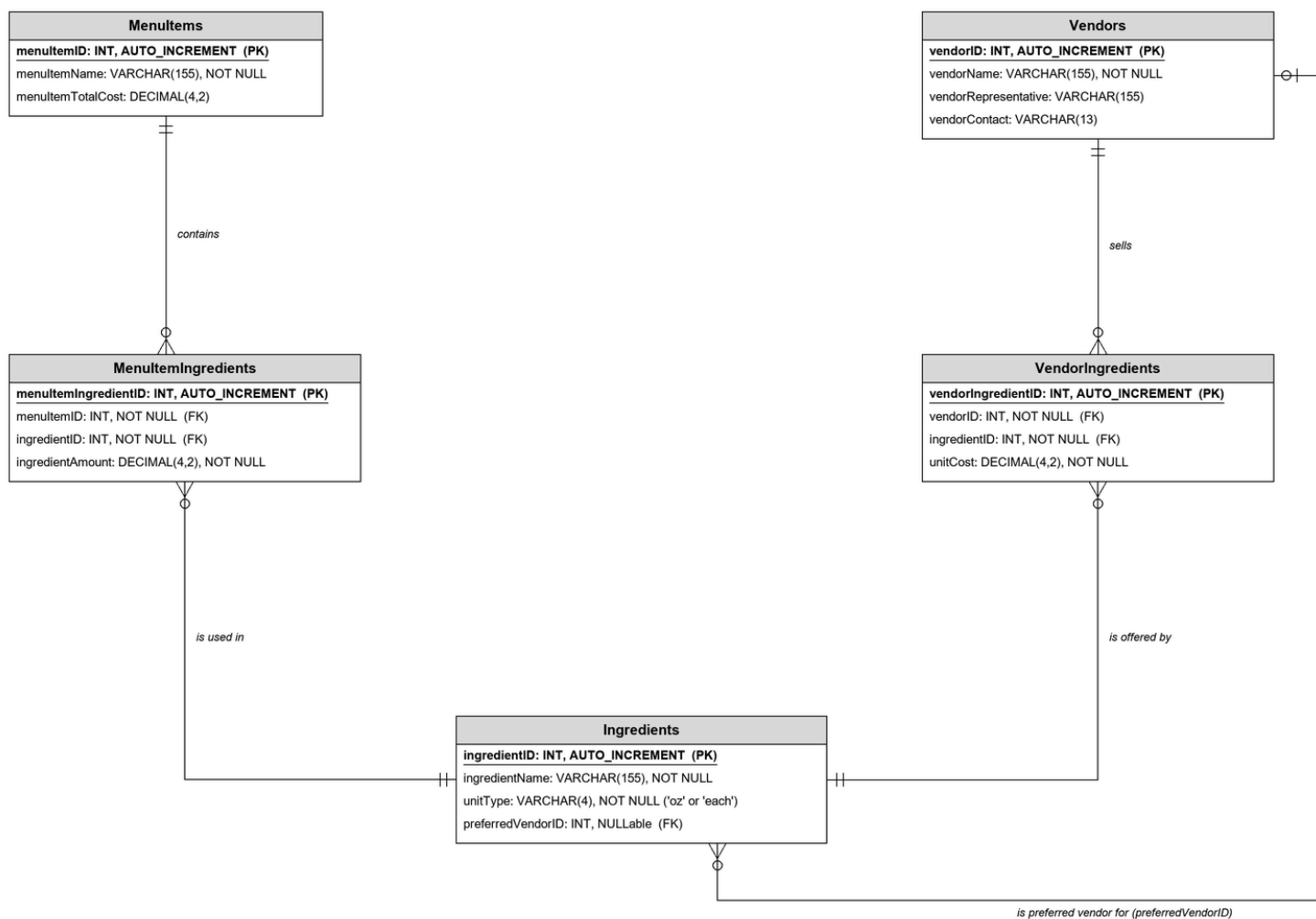
- Vendors: records the details of each vendor who sells the ingredients.
 - vendorID: int(10), auto_increment, unique, not NULL, PK
 - vendorName: VARCHAR(155), not NULL
 - vendorRepresentative: VARCHAR(155)
 - vendorContact: VARCHAR(13)
 - relationships:
 - M:N with the entity Ingredients - multiple vendors can sell multiple ingredients.
 - 1:M with the entity VendorIngredients, which serves as an intersection table.
 - 1:M with the entity Ingredients through the preferredVendorId foreign key, which identifies a preferred vendor for each ingredient based on things such as cost, quality and availability (each ingredient can have at most 1 preferred vendor at a time. A vendor can be preferred for many ingredients.)

- VendorIngredients: keeps track of the ingredients offered by each vendor, along with the corresponding cost per ingredient per unit.
 - vendorIngredientID: int(10), auto_increment, unique, not NULL, PK
 - vendorID: FK
 - ingredientID: FK
 - unitCost: decimal(4,2), not NULL
 - relationships:
 - M:1 with the entity Vendor (a vendor can sell multiple ingredients)
 - N:1 with the entity Ingredients (an ingredient can be sold by multiple vendors)

VI Entity-Relationship Diagram



VII Normalized Schema



VIII Sample Data

Table: Vendors

vendorID	vendorName	vendorRepresentative	vendorContact
1	US Foods	Sarah McGough	5419917958
2	Sysco	Marissa Zander	4583193482
3	Florence Bread Company	Gavin Smith	5419012021

Table: MenuItems

menuItemID	menuItemName	menuItemTotalCost
1	Forager Burger	7.15
2	Cod Fish & Chips	6.47
3	Siltcoos Salad	4.23
4	Steamer Clams	5.80
5	French Dip	6.04

Table: Ingredients

ingredientID	ingredientName	unitType	preferredVendorID
1	Bun	ea	3
2	Beef Patty	oz	1
3	Mushrooms	oz	2
4	Onions	oz	1
5	Smoked Gouda	oz	2
6	Lettuce	ea	1
7	Tomato	ea	2
8	Aioli	oz	2
9	French Fries	oz	2
10	Cod	oz	1

Table: MenuItemIngredients

menuItemIngredientID	menuItemID	ingredientID	ingredientAmount
1	1	1	1.00
2	1	2	5.33
3	1	3	2.00
4	1	4	2.00
5	1	5	0.75
6	1	6	1.00
7	1	7	2.00
8	1	8	2.00
9	1	9	8.00
10	2	10	8.00

Table: Vendor Ingredients

vendorIngredientID	vendorID	ingredientID	unitCost
1	1	1	0.75
2	1	2	0.53
3	1	3	0.31
4	1	4	0.25
5	1	5	0.44
6	1	6	0.15
7	1	7	0.15
8	1	8	0.15
9	1	9	0.16
10	1	10	0.63

VII Citations

Guinn, J. (2024, November 4). How to calculate restaurant COGS and boost profitability. Toast. <https://pos.toasttab.com/blog/on-the-line/restaurant-cost-of-goods-sold>

Oracle. (2026). 15.1.20.6 CHECK constraints. MySQL 8.4 Reference Manual. <https://dev.mysql.com/doc/refman/8.4/en/create-table-check-constraints.html>